IILM University- Greater Noida

Centre for Career Service

Communication Skills for Engineers II (Aptitude-Workbook)



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Chapter1 - NUMBER SYSTEM

CONVERSION FROM DECIMAL TO FRACTION

1. Convert the following into fractions:
   a. 0.777..
   b. 0.575757..
   c. 1.232323..
   d. 1.0313131..
   e. 2.05212121..
   f. $7.\overline{6}$
   g. $6.5\overline{6}$

DIVISIBILITY TEST

1. What digits should be put in place of c in 38c to make it divisible by
   a. 2 b. 3 c. 4 d. 5 e. 6 f. 9 g. 10
2. If 123X457 is divisible by 3. Find X.
3. If 1452663X is divisible by 9. Find X.
4. If 4832X18 is divisible by 11. Find X.
5. If Y7543X is divisible by 44. How many possible values of X and Y can be there?
6. If 78765XY is divisible by 12, where X and Y are two single digit numbers. How many different values for a two-digit number XY can be possible?
7. How many even integers n, where $100 \le n \le 200$, are divisible by neither 7 nor 9?
   a. 37 b. 38 c. 39 d. 40
8. 215643XY is divisible by 18, what can be the unit digit of X *Y, if both are single digit numbers?
   a. 6 b. 5 c. 1 d. 3 e. None of these
9. 306306 is always divisible by
   a. 7 b. 9 c. 11 d. 13 e. All of these
10. 555555 is always be divisible by
    a. 37 b. 5 c. 11 d. 1001 e. All of these

UNIT DIGIT

1. Find the unit digit in each of the following cases:

a. 422^{423} b. 417^{7753} c. $54^{53} \times 39^{32}$ d. 98^{96}

2. Find the unit's digit of $222^{333} + 333^{222}$.
3. Find the unit's digit of:

a. $19^{19^{19^{19}}}$.....

b. $17^{18^{19^{20}}}$ ....

4. Find the unit's digit of $8^n + 2^n$ if the unit digit of 4^n is not 6.

a. 0 b. 2 c. 4 d. 6 e. 8

5. How many distinct values can the unit digit of $1^n + 2^n + 3^n + 4^n \ldots + 9^n$ assume, where n is a natural number?

a.1 b. 2 c. 3 d. 4 e. 5

6. What will be the unit digit of $(1!)^{20!} + (2!)^{19!} + (3!)^{(18!)} + \ldots\ldots\ldots\ldots\ldots (18!)^{3!} + (19!)^{2!} + (20!)^{1!}$?

a. 0 b.1 c.3 d.5 e. None of these

7. The unit digit of the expression $28^{29} * 32^{32}$ divided by unit digit of $34^{34} * 37^{37}$. And hence answer say x is again raised the power of 372. What is the unit of of x^{372}?

a. 4 b. 2 c. 6 d. 8 e. None of these

CONCEPT OF FACTORS

Given a number N = 1280. Find the followings:

Sr.N.	Question_Type	Concept/ Formula	Answer
1.	Number of factors		
2.	Sum of factors		
3.	Number of even factors		
4.	Number of odd Factors		
5.	Number of perfect square factors		
6.	Number of perfect cube factors		
7.	Number of factors divided by 10		
8.	Sum of perfect square factors		
9.	Sum of perfect cube factors		
10.	Sum of factors divided by 10		
11.	Sum of even factors		
12	Sum of odd factors		
13.	Product of all the factors		

15. How many factors of the number $2^8 * 3^6 * 5^4 * 10^5$ are multiples of 120?

a.540 b. 660 c.594 d.792

16. Number N = $2^6 * 5^5 * 7^6 * 10^7$; how many factors of N are even numbers?

a.1183 b. 1200 c. 1050 d. 840

CONCEPT OF NUMBER OF ZEROES

1. Find the number of trailing zeroes in 137!.....
2. Find the difference between the number of zeroes in 111! and 145! ?.....
3. A number X = 128! / 5^n is given. Find the maximum value of n such that X becomes an integer. .....
4. What will be the largest value of n such that the number 153! /14^n will be the integer. .....
5. What will be the largest value of n such that the number 163!/ 20^n will be the integer. .....
6. How many zeroes will be there at the end of 5! X 10! X 15! X 20! X 25!.....
7. How many zeroes will be there at the end of 4! + 8! + 12 ! + 16! + 20!.....
8. How many trailing zeroes will be there in the number $1!^{1!}+2!^{2!}+3!^{3!}+4!^{4!}$.....$30!^{30!}$.

LCF and HCF

1. If the ratio of two numbers is 2:3 and their LCM is 54, then the sum of the two numbers is:
2. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?
3. Find the third smallest number greater than divisors which when divided by 8,10 and 12 leaves remainder 5 in each case
4. What is the least possible number which when divided by 18, 35 or 42 it leaves remainders as 2, 19 and 26 respectively?
5. Find the smallest number greater than its divisors which when divided by 9 and 11 leaves remainders 5 and 2 respectively
6. The L.C.M. of two numbers is 48. The numbers are in the ratio 2:3. Then sum of the number is:
7. Let N be the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. Then sum of the digits in N is:
8. The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively, is:
9. What is the least multiple of 13, which when divided by 5 and 7 gives remainder of 3 and 5 respectively?
10. What is the least number possible which when divided by 5 and 7 gives remainder of 3 and 2 respectively?
11. Find the greatest number that divides 150, 194 and 245, leaves remainders of 6, 2 and 5 respectively
12. Find the greatest number that divides 55, 127 and 175, leaves same remainder in each case
13. 48 rose plants, 72 marigold plants and 108 lotus plants have to planted in rows such that each row have equal number of plants and each row has plants of a particular variety only. What is the least number of rows required?
14. A rectangular cloth of (54*90) units is to be cut into equal squares. Find how many minimum no. of square pieces are required such that no cloth is wasted.
15. 6 different sweet varieties of count 32, 216, 136, 88, 184, 120 were ordered for a particular occasion. They need to be packed in such a way that each box has the same variety of sweet and the number of sweets in each box is also the same. What is the minimum number of boxes required to pack?

Chapter2 - AVERAGE

1. What is the average of the first 40 natural numbers?

   (a) 25.5 (b) 20.5 (c) 820 (d) 21

2. What is the average of the first 20 multiples of 7?

   (a) 1470 (b) 70 (c) 150 (d) 73.5

3. If the average of the sum of the first 4 consecutive Even numbers is 27. Find the largest of these numbers?

   (a) 20 (b) 24 (c) 32 (d) 30

4. If the average weight of 36 boys be 40kg and that of 44 girls be 35kg. What would be the average age of all the students?

   (a) 37.25 (b) 36.25 (c) 37.5 (d) 38.25

5. Average of 25 numbers is 18. If the average of the first 12 numbers be 14 and the average of the last 12 numbers be 17. Find the 13 th number?

   (a) 72 (b) 75 (c) 78 (d) 48

6. Average weight of Amit, Ronit, and Vijay is 45kg. Average weight of Amit and Ronit is 40kg whereas the average weight of Vijay and Ronit is 43kg. Calculate the weight of Ronit.

   (a) 31 (b) 35 (c) 42 (d) 38

7. A batsman scores 87 runs in his 17th inning, and thereby increases his average by 3 runs. Find his average after the 17th innings.

   (a) 36 (b) 39 (c) 33 (d) 42

8. The temperatures of the weekdays of a certain week starting from Sunday are respectively 20°C, 22°C, 25°C, 17°C and 24°C for the first five days. If the temperatures of the next two days are the same, and the sum of the temperatures of all the seven days of the week is 160°C, then find the temperature on Saturday.

   (a) 22°C (b) 25°C (c) 26°C (d) 28°C

9. In a town during a certain week, every day there was a 1°C, increase in temperature over the previous day. If the average temperature for the first and last days (i.e., Monday and Sunday) was 39°C, what was the average for Monday, Tuesday and Wednesday?

   (a) 37°C (b) 35°C (c) 36°C (d) 33°C

10. The average runs scored by a batsman who has played 52 innings, was increased by one after an innings of 126. Find the average runs of the batsman before this innings.

    (a) 71 (b) 73 (c) 75 (d) 74

11. The average age of a group went up by 2 years when a man aged 34 years was replaced by an old man aged 58 years. How many members were there in the group?

    (a) 13 (b) 15 (c) 12 (d) 16

12. The average marks of the boys and the girls in a class are 30 and 40 respectively. Find the average marks of all the students in the class.

    (a) 20 (b) 40 (c) 30 (d) cannot be determined

13. The average marks scored by Monika in five out of six subjects in an exam is 60. If Monika scores an average of 75 marks in the six subjects, find his score in the sixth subject.

    (a) 125 (b) 160 (c) 150 (d) 120

14. The average age of 10 oarsmen increase by 1.8kg, when one crew member who weighs 53kg is replaced by a new man. What is the weight of the new person who joined the crew.

    (a) 54.8kg (b) 71kg (c) 51.2kg (d) 28kg

15. In a class two students aged 16 years and 18 years are replaced by students who are aged 19 years and 21 years. If the average age of the class increased by 3 months, then how many students are there in the class

    (a) 20 (b) 24 (c) 22 (d) 23

16. The average weight of certain number of boys in a group is 30 kg. If 5 boys with an average weight of 12 kg join the group, the average weight would be the same as if 5 boys with an average weight of 36 kg leave the group. Find the number of boys in the group.

    (a) 10 (b) 15 (c) 12 (d) 14

17. 9 friends went out for a meal. 8 of them spent Rs12 each on their meals, while the 9th person spent Rs 8 more than the average expenditure of all the 9 people. How much was spent in total behind the meal.

    (a) 116 (b) 128 (c) 117 (d) 113

18. There were 35 students in a hostel. Due to the admission of 7 new students, the expenses increased by Rs42/day, while the average expenses per head went down by Re1. What was the original expense of the students?

    (a) 420 (b) 740 (c) 350 (d) 462

19. The average age of a group of students is 16 years. If a 66-year-old teacher joins the group, the average increases by one year. The number of students in the group is

    (a) 48 (b) 49 (c) 52 (d) 51

20. The average age of 30 students of a class is 16years. If the age of the teacher is also included, the average increased by 1 year. Find the age of the teacher

    (a) 47 (b) 43 (c) 45 (d) 49

Chapter3 - CLOCK

1. A clock is started at 06:00 in the morning. Through how many degrees will the hour hand rotate when the clock shows 11:00 in the morning?

   (a) 140 degree (b) 180 degree (c) 150 degree (d) 160 degree

2. At 06:15 the hour hand and the minute hand of a clock form an angle of

   (a) 90 degree (b) 91.5 degree (c) 97.5 degree (d) 94.5 degrees

3. What will be the angle (in degree) between the hands of the clock when the time is 5:40?

   (a) 70 (b) 60 (c) 55 (d) 80

4. What will be the time between 6 and 7, when the angle is 70 degree?

   (a) 6:20 (b) 6:45 (c) 6:22 (d) 7:00

5. The reflex angle between the hands of a clock at 10:25 is

   (a) 180.5 degree (b) 162.5 degree (c) 165.5 degree (d) 197.5 degree

6. At what time between 5 p.m. and 6 p.m., do the hands of a clock coincide?

   (a) 6 hour 29 (3/11) min (b) 5 hour 33 (8/11) min (c) 5 hour 246/11 min (d) 5 hour 27 (3/11) min

7. Find at what time (in minutes) past 8 o' clock but before 9 o' clock will the hands of a clock be in the same straight line but not together.

   (a) 100/11 (b) 110/11 (c) 120/11 (d) None of these

8. How many times do the hands of a clock coincide in a day?

   (a) 44 (b) 22 (c) 11 (d) 24

9. How many times do the hands of a clock at right angle in a day?

   (a) 44 (b) 22 (c) 11 (d) 24

10. If a wall clock shows 9 h 30 min. What time will the clock show in the mirror?

    (a) 2 h 45 min (b) 3 h 30 min (c) 2 h 30 min (d) None of these

11. How much does a watch lose per day if its hands coincide every 64 minutes?

    (a) 32 and (8/11) min (b) 36 and (5/11) min (c) 32 and (5/11) min (d) 30 min

12. The number of minutes from midnight to now is 9 times the number of minutes from now to noon. What time is it now?

    (a) 10:50 AM (b) 10:48 AM (c) 10:40 AM (d) 10:53 AM

13. A clock is set right at 10 am. The clock gains 10 min in a day. What will be the true time when the watch indicates 3 pm the next day?

(a) 12 min past 2 PM (b) 45 min past 2 PM (c) 48 min past 2 PM (d) 30 min past 2 PM

14. A clock is set to show the correct time at 7 AM Monday. The clock loses 15 min in 24 h. What will be the true time when the clock indicates 6 AM on the following Friday?

(a) 6: 15 AM Friday (b) 6: 30 AM Friday (c) 7 AM Friday (d) 7: 15 AM Friday

15. My watch was 8 minutes behind at 8 PM on Sunday but it was 7 minutes ahead of time at 8 PM on Wednesday. During this period, at what time this watch has shown the correct time?

(a) Tuesday 10:24 AM (b) Wednesday 9:16 PM (c) Tuesday 10:24 PM (d) Wednesday 9:16 AM

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Chapter4 - CALENDAR

1. If the 5th day of a month is Friday, which of the following will be the 7th day from 10th of that month?
   (a) Tuesday (b) Monday (c) Wednesday (d) Thursday
2. Today is Thursday. The day after 59 days will be:
   (a) Friday (b) Thursday (c) Saturday (d) Sunday
3. The last day of a century cannot be: (a) Monday (b) Wednesday (c) Tuesday (d) Friday
4. How many leap years do first 300 years have? (a) 75 (b) 74 (c) 72 (d) 73
5. It was Tuesday on Feb 8, 2005. What was the day of the week on Feb 8, 2004?
   (a) Monday (b) Thursday (c) Friday (d) Sunday
6. 8th Dec 2009 was Tuesday, what day of the week was it on 8th Dec, 2006?
   (a) Sunday (b) Tuesday (c) Friday (d) Tuesday
7. If 10th May, 1997 was a Monday, what was the day on Oct 10, 2001?
   (a) Saturday (b) Sunday (c) Thursday (d) Friday
8. What was the day of the week on December 26, 1995?
   (a) Friday (b) Tuesday (c) Sunday (d) Monday
9. What was the day on 31st October 1984? (a) Friday (b) Sunday (c) Wednesday (d) Monday
10. On what days of July, 1976 did Sunday fall?
    (a)1st, 8th, 16th, 23rd (b) 6th, 20th, 27th (c) 4th, 11th, 18th, 25th (d) 14th, 21st, 28th
11. Which two months in a year have the same calendar?
    (a) October, December (b) April, November (c) June, October (d) April, July
12. Find the year from 2015 to 2030 which have the same calendar as that of 2015.
    (a) 2019 (b) 2023 (c) 2026 (d) 2029
13. The year next to 1990 will have the same calendar as that of the year 1990 is
    (a) 2001 (b) 1997 (c) 1996 (d) 1992
14. How many times the 29th day of the Month does occur in 400 consecutive years?
    (a) 4500 (b) 4498 (c) 4497 (d) 4495
15. Pinky was born on 29th, Feb 2016 which happened to be a Monday. If she lives to be till 2099, how many birthdays would she celebrate on a Monday?
    (a) 1 (b) 2 (c) 3 (d) 5

16. If we suppose the 60th Independence Day of India was on Thursday, then the 85th Independence Day would have been on?

    (a) Monday (b) Wednesday (c) Friday (d) Sunday

Chapter5 - PERCENTAGE

1. What is the value of 37.5 % of 488?

   (a) 190 (b) 183 (c) 170 (d) None of These

2. If 40% of a number exceeds 25% of the same number by 105. What is the value of the number?

   (a) 161 (b) 1050 (c) 700 (d) 690

3. Two numbers are less than a third number by 30 % and 37% respectively. How much percent is the second number less than the first?

   (a) 10% (b) 15% (c) 20% (d) 25%

4. If the salary of Raman has increased by 20 %, then by what percentage the new salary has to be decreased so that it becomes equal to the original salary?

   (a) 20% (b) 16.66% (c) 25% (d) 14.28%

5. A positive number is divided by 5 instead of being multiplied by 5. What percent is the result of the required correct value?

   (a) 4% (b) 25% (c) 5% (d) 10%

6. Because of scarcity of rainfall, the price of a land decreases by 12% and its production also decreases by 4%. What is the total effect on revenue?

   (a) Loss of 16% (b) Loss of 15.48% (c) Gain of 15.48% (d) Loss of 15.52%

7. From the salary of an officer, 10% is deducted as house rent, 20% of the rest, he spends on conveyance, 20% of the rest, he pays on income tax and 10% of the balance, he spends on clothes. Then, he is left with Rs. 15552. Find his total salary?

   (a) Rs.25000 (b) Rs. 30000 (c) Rs. 35000 (d) Rs. 40000

8. After deducting 10 % from a certain sum, and then 20 % from the remainder, Rs 3600 is left. Find the original sum?

   (a) 5000 (b) 2592 (c) 4500 (d) 3960

9. A, B, C are 3 students. A got 20% more marks than B and 30 % less than C. if B got 175, the how much C got?

   (a) 175 (b) 210 (c) 300 (d) None of These

10. 40 liter of a mixture of milk and water contains 10 % water. How much water must be added to make the water 20 % in the new mixture?

    (a) 5 (b) 4 (c) 10 (d) None of These

11. Reduction in price of sugar by 20% allows a household to buy 45 kg more for Rs.450. Find the original price of the sugar.

    (a) Rs 2 kg (b) Rs 2.5 kg (c) Rs 3 kg (d) None of These

12. A candidate scores 25 % marks and fails by 30 marks, while another candidate who scores 50 % marks get 20 marks more than the minimum marks required to pass the examinations. Find the maximum marks for the examination.

    (a) 200 (b) 100 (c) 400 (d) None of These

13. Forty percent of the employees of a certain company are men, and 75 percent of the men more than Rs. 25,000 per year. If 45 percent of the company's employees earn more than Rs. 25,000per year, what fraction of the women employed by the company earn Rs. 25,000 year or less?

    (a)3/4 (b)9/20 (c) 11/20 (d) 1/4

14. In an election between two candidates, 75% of the voters cast their votes, out of which 2% of the voters were declared in-valid. A candidate got 9261 votes which were 75% of the total valid votes. The total number of voters enrolled in that election was :

    (a) 16000 (b) 16400 (c) 16800 (d) 18000

15. A got 37.5 % marks less than B, then by what percent the marks of B is more than the marks of A?

    (a) 60 % (b) 37.5 % (c) 27.27 % (d) None of these

16. If the numerator of a fraction is increased by 250% and the denominator is increased by 400%, the resultant fraction is7/19.What is the original fraction?

    (a) 10/19 (b) 56/95 (c)9/5 (d) 19/7

17. Rs. 395 are divided among A, B and C in such a manner that B gets 25% more than A and 20% more than C. The share of A is?

    (a) Rs. 198 (b) Rs. 120 (c) Rs. 180 (d) Rs. 195

18. The production of a company increases by 4%, 5% and 6% successively in three successive years, then what is the ratio of the final to the initial production?

    (a) 1.15 (b) 15.752 (c) 1.15752 (d) 2.5752

19. By how much percentage the side of the square should be increased so that the area of the square increases by 21 %?

    (a) 21% (b) 10% (c) 11% (d) Can't say

20. If the radius of the cylinder increases by 10 % and the height increases by 20 %. Then, what is the change in the volume of the cylinder?

    (a) 40% (b) 32% (c) 21% (d) 45.2%

21. A student was asked to measure the length and breadth of a rectangle. By mistake, he measured the length 20% less and the breadth 10% more. If its original area is 200 sq cm, find the area after this measurement?
    (a)176 sq cm (b) 206 sq cm (c) 226 sq cm (d) 316 sq cm
22. Sohan spends 23% of an amount of money on an insurance policy, 33% on food; 19% on children's education and 16% on recreation. He deposits the remaining amount of Rs. 504 in bank. How much total amount does he spend on food and insurance policy together?
    (a) 3200 (b) 3126 (c) 3136 (d) 3048
23. Sameer spends 24% of his monthly income on food and 15% on the education of his children. Of the remaining salary, he spends 25% on entertainment and 20% on conveyance. He is now left with Rs. 10736. What is the monthly salary of Sameer?
    (a) 27600 (b) 28000 (c) 31200 (d) 32000
24. A reduction of 20% in the price of wheat enables a person to buy 3.5 kg more wheat for Rs. 770. The original price of wheat is?
    (a) Rs. 55 kg (b) Rs. 45 kg (c) Rs. 65 kg (d) Rs. 37 kg
25. In a class of 40 students and 5 teachers, each student got sweets that are 15% of the total number of students and each teacher got sweets that are 20% of the total number of students. How many sweets were there?
    (a) 280 (b) 240 (c) 320 (d) 360
26. In an examination, it is required to get 296 marks out of aggregate marks to pass. A student gets 222 marks and is declared failed by 10% marks. What are the maximum aggregate marks a student can get?
    (a) 830 (b) 810 (c) 780 (d) 740
27. A statue of 80 kg contains 60% copper, 15% zinc and remaining aluminium. After melting if 5% aluminium is removed, find the quantity of aluminium required to make the statue?
    (a) 21 1/19 kg (b) 17 5/29kg (c) 19 kg (d) 21 kg
28. A 40 liters mixture of alcohal and water contains 10 % alcohal. How much water should be added so that the alcohal becomes 5% of the total mixture?
    (a) 40 (b) 20 (c) 4 (d) 80
29. Directions: Refer to the following information to answer the questions that follow.
    In an examination, a candidate who secures 25% of the maximum marks fails by 60 marks, but another candidate who secures 42% gets 8 marks more than that necessary to pass.
    The maximum marks is?
    (a) 400 (b) 600 (c) 1000 (d) 420
    The pass marks is?
    (a) 660 (b) 160 (c) 400 (d) 260

The pass percentage is?

(a) 35% (b) 33% (c) 38% (d) 40%

30. 8% of the voters in an election did not cast their votes. In the election, there were only two candidates. The winner by obtaining 48% of the total votes in the voters list defeated his contestant by 1100 votes. The total number of voters in the election was?

    (a) 21000 (b) 22000 (c) 23500 (d) 27500

31. In a group of persons, 70% of the persons are male and 30% of the persons are married. If two-sevenths' of the males are married, what fraction of the females is single?

    (a) 2/7 (b) 1/3 (c) 3/7 (d) 2/3

32. The tank-full petrol in Arun's motor-cycle lasts for 10 days. If he starts using 25 % more every day, how many days will the tank full petrol last?

    (a) 5 (b) 6 (c) 7 (d) 8

33. Amar, Akbar and Anthony are friends, being looked after by a matron Farah. Amar weighs 50% more than Akbar and Anthony weighs 25% less than Amar. Farah weighs a third of the combined weight of the three boys. All four together weight 232 kg. The correct arrangement of the person in the ascending order of their weights is

    a. Anthony, Akbar, Farah, Amar

    b. Anthony, Akbar, Amar, Farah

    c. Akbar, Anthony, Amar, Farah

    d. Akbar, Anthony, Farah, Amar

34. In a factory, 40% of the staff is female. 40% of the female and 60% of the male voted for me. The percentage of vote I got was?

    (a) 24 (b) 42 (c) 50 (d) 52

35. A city has a population of 3,00,000, out of which 1,80,000 are males. 50% of the population is literate. If 70% of the males are literate, the number of literate females is?

    (a) 24,000 (b) 30,000 (c) 54,000 (d) 60,000

36. The number of votes not cast for the Praja Party increased by 25% in the National General Elections over those not cast for it in the previous Assembly polls, and the Praja party lost by a majority twice as large as that by which it had won the Assembly Polls. If a total of 2,60,000 people voted each time, how many voted for the Praja Party in the previous Assembly Polls?

    (a) 1,10,000 (b) 1,50,000 (c) 1,40,000 (d) 1,20,000

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Chapter6 - PROFIT LOSS AND DISCOUNT

Type 1

1. If I buy @ 3 articles in Rs. 5 and sells @ 5 articles in Rs. 8. Find the profit or loss percentage.
2. If I buy @ 30 articles per rupee, how many for a rupee should sell to gain 20%?
3. A man bought oranges at the rate of 8 for Rs 34 and sold them at a rate of 12 for Rs 57.How many oranges should be sold to earn Rs 45 as a profit.

Type 2

4. CP of 40 articles is equal to SP of 50 articles. Find the profit or loss percentage.
5. On buying and selling 30 apples, a merchant made a profit equals to the SP of 6 apples. Find the profit percentage.
6. Cost price of 12 oranges is equal to selling price of 9 oranges and the discount on 10 oranges is equal to the profit on 5 oranges. What is the percentage difference between the profit percentage and discount percentage?

Type 3

7. One article is sold at 12.5% profit and another at 16.66% loss. Find the overall profit or loss percentage.
   (i) If the CP of both the articles are same.
   (ii) It the SP of both the articles are same.
   (iii) If the CP of first article is same as the SP of second article.
   (iv) If the SP of first article is equal to the CP of second article.

Type 4

8. Article sold at a certain price result in 20% profit. If sold at Rs. 300 less, It would have been resulted in 10 % loss. What would be the SP to gain 25%?
9. An article costing Rs. 250 when sold at a certain price results in a gain of X%. Had the Cost Price been Rs. 50 less and the Selling Price been Rs. 100 more, it should have been resulted in a gain of 2X%. Find X.
10. Profit earned when a person sells an article at Rs 370is 20% more than the loss incurred when he sells the article at Rs. 315. Find the cost price.
11. A man sells an article at a profit of 20%. If he had bought it at 10% less and sold at Rs. 18 more, he would have been made 40% profit. Find there CP

Type 5

12. On allowing a discount of 10% on an article. The shopkeeper gains 20%. If a discount of 20% is allowed on it, what will be the gain percentage?

13. A shopkeeper earns a profit of 12% on selling a book at 10% discount on the printed price. The ratio of the cost price to the printed price is?
14. The marked price of an electric rod is Rs. 690. The shopkeeper allows a discount of 9.09% and gains 11.11%. If no discount is allowed his gain percent would be?
15. A trader lists his articles 6.66% above the cost price and allows a discount of 25% on cash payment. His loss percent is?

Type 6

16. A manufacturer sells an article to a dealer at 12.5% profit. The dealer sells the article to a customer at 16.66% profit. If the customer pays Rs. 210 for the article, what did the article cost to the manufacturer?
17. A shopkeeper allowed two successive discounts of 50% and 60% on an article. Find the net change in the price of the article after discounts.

Type 7

18. A trader claims to sell his goods at the same rate as his purchase rate. However, his balance reads 1 kg for 900 gms. Find his profit percentage.
19. A trader sells his goods at 10% mark-up over his purchase price. However, he uses a faulty half kg weight that weighs 600 gms. Find his profit percentage.
20. In summers, a meter scale expands by 10%. What will be the profit percentage of a cloth trader, who sells at a mark up of 20% over his purchase price?

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Chapter7 - SI and CI

Simple Interest

1. If A deposits Rs. 20,000 in a bank for 3 years at a rate of 10%, what is the simple interest he will get at the end of the period?
2. Sagar borrows Rs. 50,000 from a bank for 5 years. What is the rate of simple interest charged by the bank if after 5 years Sagar had to pay Rs. 66,000 to the bank?
3. After how many years would an amount double itself at 15% rate of simple interest?
4. An amount quadruples in 20 years. Find the rate per anum
5. The rate of simple interest of a bank is 10%. Akash was taking a loan of a huge amount for one year so the bank agreed to give him the loan at a rate of 8%. They reasoned that even with 8% they will get twice the interest that they would have got had they given out a loan of Rs. 40,00,000 at 10% for a year. What is the amount of Akash's loan?
6. Ajay takes a loan of Rs. 30,000 from a bank for 8 years at 6.5% rate of simple interest. He then loans out Rs. 20,000 for 8 years at 7.5% rate of simple interest. He could loan out the balance only at 5.5% for 8 years. Inthe entire transaction, did Ajay make or lose money and how much?
7. The underworld's don Chhota Chatri loans money to people at simple interest. He charges a certain rate of interest for the first year. Next year he doubles the initial rate of interest on the amount. Third year he triples the initial rate of interest and so on… A man took an amount of Rs. 9000 and after 3 yrs paid back anamount of Rs. 15000 back. What was the rate of interest in the first year?
8. I needed Rs. 1,20,000 to buy a Plasma TV and hence I borrowed Rs. 75,000 from Vani and the rest from Vivek. Vani and Vivek charge me a rate of interest such that the interest amount payable to both of them is the same. If in all I re-pay them a total of Rs. 1,50,000 at the end of 2 years, what is the rate of interest charged per annum by the two?
9. Vijay took part of Rs. 10,000 loan at 4% and the rest at 6%. If he pays a total interest of Rs. 900in two years,find the amount taken on loan at 4%. The interest rate charged by the bank is Simple Interest.
10. A cooperative bank lent Rs.4000 to Anand at a certain rate of simple interest and Rs.5000 to Milind at ½% more than that of Anand. After 2 years the bank received Rs. 860 as total interest from Anand and Milind. Find the rate of interest per annum at which the amount was lent to Milind.

Compound Interest

11. A certain sum of money under compound interest becomes 3 times of itself in 12 years. Find
    (I) Rate of Interest
    (II) In what time it will be 243 times of itself

12. At what rate of compound interest per annum will a sum of Rs. 1200 become Rs. 1348.32 in 2 years?
13. A sum of money invested at CI amounts to Rs. 2400 in 3 years and 2520 in 4 years. Find the rate of interestper annum?
14. A certain sum amounts to Rs. 8000 in 2 years and 27000 in 5 years. Find the principal?
15. In how many years, a certain sum of money of Rs. 1,53,600 amounts to Rs. 2,18,700, if the rate of interest is 12.5% p.a.c.a. ?
16. Difference between CI and SI = Rs. 20 Rate = 4% p.a.c.a. Time = 2 Years Principal = ?
17. Difference between CI and SI = 4300 Rs. Rate = 7.14% p.a.c.a. Time = 3 Years, Principal = ?
18. Compound interest for 3rd year = 1980Rs. Rate = 11.11% p.a.c.a. Time = 3 Years Principal = ?

PRACTICE QUESTIONS

1. Ravi Shankar takes money from the employees' cooperative society at lower rate of interest and saves in a scheme, which gives him a compound interest of 20%, compounded annually. Find the least number of complete years after which his sum will be more than double.
2. Equal sums of money are deposited in two different banks by M/s Enterprises, one at compound interest (compounded annually) and the other at simple interest, both at 5% per annum. If after two years, the difference in the amounts comes to Rs. 200, what are the amounts deposited with each bank?
3. An amount of 7500 is invested in a compound interest scheme for 4 years. The rate of interest is 2% for the first year, 3 % for next 2 years, and for the last year, it is 4%. The final amount is
4. Rs. 5887 is divided between Shyam and Ram, such that Shyam's share at the end of 9 yr is equal to Ram's share at the end of 11 yr, compounded annually at the rate of 5%. The share of Shyam is?
5. The population of Dehradun was 15000 two years ago. If it had increased by 2% and 3% in the last two years, find the present population of Dehradun
6. Mr X, a very industrious person, wants to establish his own unit. For this he needs an instant loan of Rs 5,00,000 and, every five years he requires an additional loan of Rs. 100,000. If he had to clear all his outstandings in 20 yrs, and he repays the principal of the first loan equally over the 20 yrs, find what amount he would have to pay as interest on his initial borrowing if the rate of interest is 10% p.a. SI ? A)5,60,000 B) 5,40,000 C) 5,50,000 D) 5,00,000
7. If money is invested at r percent interested, compounded annually, the amount of the investment will double in approximately 70/r years. If Pat's parents invested $5,000 dollars in a long-term bond that pays 8 percent interest, compounded annually, what will be the approximate total amount of the investment 18 years later, when Pat is ready for college?

   A. 20,000 B. 15,000 C. 12,000 D. 10,000 E. 9,000
8. An investment of $1000 was made in a certain account and earned interest that was compounded annually. The annual interest rate was fixed for the duration of the investment, and after 12 years the $1000 increased to $4000 by earning interest. In how many years after the initial investment was made the $1000 have increased to $8000 by earning interest at that rate?

   A. 16 B. 18 C. 20 D. 24 E. 30
9. On July 1, 1982, Ms. Fox deposited $10,000 in a new account at the annual interest rate of 12 percent

compounded monthly. If no additional deposits or withdrawals were made and if interest was credited on the last day of each month, what was the amount of money in the account on September 1, 1982? A. $10,200 B. $10,201 C. $11,100 D. $12,100 E. $12,544

10. Some amount was lent on 6% per annum simple interest. After one year, Rs.6800 is repaid and the rest of the amount is repaid at 5% per annum. If the second year's interest is 11/20 of the first year's interest, find what amount of money was lent out.

    A. 17000 B. 16800 C. 16500 D. 17500 E. None of these

11. A is going to pay Rs 836 to B 9 months later at 6% annual simple interest. B is going to pay A Rs 783, 15 months later later at 7% annual simple interest. If they decided to settle the debts right now, who will pay what amount to whom?

    A. A pays Rs. 80 to B right now B. B pays Rs. 80 to A right now

    C. A pays Rs. 160 to B right now D. B pays Rs. 160 to A right now

12. Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?

    A. Rs. 6400 B. Rs. 6500 C. Rs. 7200 D.Rs. 7500

13. A sum of Rs. 725 is lent in the beginning of a year at a certain rate of interest. After 8 months, a sum of Rs. 362.50 more is lent but at the rate twice the former. At the end of the year, Rs. 33.50 is earned as interest from both the loans. What was the original rate of interest?

    A. 3.6% B.4.5% C.5% D.6%

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Chapter8 – Direction Sense

1. After going 60m towards North, Tom takes a right turn and goes for another 16m.he then takes a right turn and goes 48m.How far is he from starting point?

   a)20m b)12m c)16m d)24m

2. Y is in the East of X which is in the North of Z. If P is in the South of Z, then in which direction of Y, is P

   a) North b) South c) South-East d) None of these

3. Beena starts at point T, walks straight to point U which is 4 m away. She turns left at 90° & walks to W which is 4m away. Turns 90° right & goes 3m to P, turns 90° right & walks 1m to Q turns left at 90° & goes to V which is 1m away & once again turns 90° right & goes to R 3m away. What is distance between T & R?

   a) 4 m b)5m c) 7m d) 8m e) cannot be determined

4. A watch is so placed that at 4 pm the minute hand points towards north-west.in which direction does the hour hand points at 7 pm?

   a) north-east b) south-east c) north- west d) south-west

5. One morning after sunrise, Sita and Geeta were standing at gandhimarg road in Mumbai with their back towards each other. Sita's shadow fell exactly right hand side. Which direction was Geeta facing?

   a) east b) west c) north d) south

6. Rashmi started from her house and walked 20m towards east, where her brother Naitik joined her. Both of them turned left and walked 20m.There was a quarrel between the two and got separated. Rashmi turned right and her brother turned left. Rashmi walked 20m and turned left. Naitik walked 30m and then turned left. Rashmi walked 10m to the left. Finally, she turned left again, walked 6m and reached her friend's house. Naitik who turned left walked 10m and finally he turned left, walked 4m and reached his friend's house. Which direction is Naitik facing now?

   a) west b) east c) north d) south

7. City M is situated east of city N. City O is situated south of city N. City P is situated north of city O.in which direction city N is situated with respect to city P?

   a) north b) south c) east d) CND

8. A, B, C, D, E, F, G, H & I are 9 houses. C is 2km east of B. A is 1km North of B & H is 2km South of A. G is 1km West of H while D is 3km East of G & F is 2km North of G. I is situated just in middle of B & C while E is just in middle of H & D. So distance between A & F is?

   A) 1km B) 1.41km C) 2km D) 3km E) Cannot be determined

9. Village Chimur is 20km to the North of Village Rewa. Village Rahate is 18km to the East of Village Rewa. direction Village Angne is 12km to the west of Chimur. If Sanjay starts from Village Rahate and goes to Village Angne in which is he from his starting point?

   A) North B) North- West C) South D) South- East

10. B is to the South-West of A, C is to the East of B and South East of A, D is to the North of C in the line with B and A.D is in which direction, with respect to A?

    A) North B) East C) South- East E) North- East

11. A watches Read 4.30. If the minute hand points east. In which direction will the Hour hand point?

    A) South - East B) North - East C) North D) North- West

12. Ms. A goes for her morning walk at 6 'O' clock towards Sun for 2km. Then she turns to her right and walks 3km. She again turns to her left and walks 2km. Finally, she turns to her left to walk another 6km. In which direction is she moving and at what distance from the last turn she is standing?

    A) 6km East B) 9km East C) 6km North D) 9km North

13. A Postman was returning to the Post Office which was front of him to the North. When the Post Office was 100m away from him, he turns to the left and moved 50m to deliver the LAST letter at the Shanti Villa. He then moved in the same direction for 40m turned to his right and moved 100m. How many meters was the away from the Post Office?

    A) 0 B) 150 C) 90 D) 100

14. Neeraj wants to go to School from his house. First of all, he goes to the crossing from here he turns to the right and reaches the bus stand. Bus stand is opposite to the library. In which direction is School located?

    A) North B) East C) Cannot be determined D) West

15. If South- East is called 'East', North- West is called 'West', South West is called 'South' & so on what will North be called?

    A) East B) South C) North East D) North West E) None of these

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Chapter9 - BLOOD RELATION

1. A family has a man, his wife, their four sons and their wives. The family of every son also has 3 sons and one daughter. Find out the total number of male members on the whole family?

   (a) 4 (b) 8 (c) 12 (d) 17

2. A and B are brothers. C and D are sisters. A's son is D's brother. How is B related to C?

   (a) Father (b) brother (c) Grandfather (d) Uncle

3. A is B's sister. C is B's mother. D is C's father, E is D's mother. Then, how is A related to D?

   (a) Grandmother (b) Grandfather (c) Daughter (d) Grand daughter

4. Read the following information carefully and answer the questions given below: Ravi is the son of Aman's father's sister. Sahil is son of Divya who is mother of Gaurav and grandmother of Aman. Ashok is the father of Tanya and grandfather of Ravi. Divya is wife of Ashok.

   (i) How is Ravi related to Divya?

   (a)Nephew (b) Grandson (c) Son (d) Data inadequate

   (ii) How is Gaurav's wife related to Tanya?

   (a)Niece (b) Sister (c) Sister-in-law (d) Data inadequate

5. Read the following information carefully and answer the questions given below: There are six children playing football, namely A, B, C, D, E, and F. A and E are brothers. F is the sister of E. C is the only son of A's uncle. B and D are daughter of the brother of C's father.

   (i)How is C related to F?

   (a) Cousin (b) brother (c) son (d) Uncle

   (ii)How many male players are there?

   (a) One (b) Three (c) Four (d) Five

   (iii)How many female players are there?

   (a) One (b) Two (c) Three (d) Four

   (iv)How is D related to A?

   (a) Uncle (b) Sister (c) Niece (d) Cousin (e) Can't Say

6. Pointing towards A, B says that "Her mother is the only daughter of my mother". How is B related to A?

a) Can't Say b) Aunt c) Daughter d) Mother

7. Introducing a boy Mohan says, "The brother of this boy's mother, is the only son of my mother's father." How is the mother of the boy related to Mohan?

   a) Grand mother b) Either Aunt or Mother c) Mother d) Aunt

8. Pointing to a woman in the photograph A said, "She is the daughter of the daughter of the only son of my paternal grandfather." How is the woman related to A?

   a) Sister b) Maternal aunt c) Niece d) Cousin

9. Pointing to Mona, Vipul said, "She is the wife of the only son of my father's father." How is Mona related to Vipul?

   a) Mother b) Sister c) Aunt d) Sister-in-law

10. Pointing to a man in photograph Monalisha says, "The father of his brother is the only son of my maternal grandfather." How is the man related to Monalisha?

    a) Husband b) Son c) Cousin d) Father

11. If 'A$B' means 'A is the father of B', 'A * B' means 'A is the mother of B' 'A @ B' means 'A is the wife of B', then which of the following means 'M is the grandmother of N'?

    (a) M*T$N@R (b) M*T$R@N (c) M*R$T@N (d) M*R@T@N

12. Read the following information carefully to answer the question given below:

    'A × B' means 'A is the brother of B'; 'A ÷ B' means 'B is the father of A'; 'A + B' means 'A is the sister of B'; 'A - B' means 'A is the mother of B';

    (i) Which of the following means 'Q is the paternal uncle of K'?

    (X) K × P ÷ M × Q

    (Y) K × B ÷ N × Q × D

    (Z) Q × L ÷R × K

    (a) Only (X) (b) Only (Y) (c) Only (Z) (d) Both (X) and (Y) (e) None of these

    (ii) Which of the following statements is/are superfluous to answer the above question?

    (a) (i) only (b) (ii) only (c) (iii) only (d)Only (i) and (iv) (e) None

13. If 'P+Q' means 'P is the father of Q' 'P×Q' means 'P is the brother of Q'; 'P-Q' means 'P is the mother of Q', then which of the following is definitely true about C-A+B?

    (a) B is the son of A. (b) A is the son of C. (c) B is the father of A. (d) C is the mother of B.

14. If X is the brother of the son of Y's son, how is X related to Y?

    a) son b) brother c) cousin d) grandson